Data Engineering Pipeline for E-Commerce Sales Analytics: A Case Study Using Amazon India Sales Dataset

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Abstract—This paper presents the design and implementation of a Data Engineering Pipeline for analyzing e-commerce sales data from Amazon India (April–June 2022). The pipeline integrates MySQL for raw data storage (OLTP layer), Pentaho Data Integration (PDI) for Extract-Transform-Load (ETL) processing, and Microsoft Power BI for interactive visualization. The dataset comprises 128,975 transaction records covering 24 attributes including order information, product categories, customer locations, and sales amounts. Through a Star Schema-based Data Warehouse model comprising one fact table (SalesFact) and five dimension tables (CustomerDim, ProductDim, LocationDim, StatusDim, TimeDim), the processed data enables multi-dimensional analytical queries. Key findings reveal total revenue of INR 78.59 million across 120,378 unique orders, with the Set and Kurta categories dominating at 67% of revenue. Maharashtra and Karnataka emerged as the top markets, while April registered peak monthly revenue. The layered architecture—Raw Layer, Staging Layer, and Data Warehouse Layer—demonstrates a scalable approach for retail analytics. This study contributes a reproducible reference pipeline for academic and business intelligence practitioners.

Keywords—data engineering pipeline; ETL; data warehouse; star schema; e-commerce analytics; Pentaho; MySQL; Power BI; Amazon sales

I. INTRODUCTION

The rapid growth of e-commerce has generated massive volumes of transactional data that, when properly harnessed, provide actionable insights for strategic decision-making [1]. Platforms such as Amazon generate millions of order records daily, containing rich information about customer behavior, product demand, geographic distribution, and seasonal trends. However, the raw form of these records—typically flat CSV exports—is not optimized for analytical queries and requires significant transformation before it can yield meaningful business intelligence [2].

Data Engineering, an emerging sub-discipline of data science, addresses this challenge through the systematic construction of pipelines that ingest, clean, transform, and serve data to downstream analytics consumers [3]. A canonical pipeline typically traverses three logical layers: an Operational Data Store or raw layer where source data lands unchanged, a staging or cleansed layer where data quality rules are applied, and a warehousing layer structured for analytical workloads [4].

This paper describes the end-to-end implementation of such a pipeline, applied to the Amazon India Sales dataset sourced from Kaggle, covering the period from March to June 2022. The chosen technology stack—MySQL for relational storage, Pentaho Data Integration (PDI) for ETL orchestration, and Power BI for dashboarding—represents a widely adopted open-source-friendly toolkit accessible to academic teams and small-to-medium enterprises [5][6].

The primary contributions of this work are: (1) a reproducible three-layer data architecture (OLTP → Staging → OLAP) for retail analytics; (2) a comprehensive ETL workflow in Pentaho that handles missing value imputation, type coercion, and row-level validation; (3) a Star Schema design comprising five dimension tables and one fact table, together with its physical implementation in MySQL; and (4) a Power BI dashboard that surfaces six key performance indicators and four revenue decomposition views.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the dataset. Section IV presents the system architecture. Section V details the ETL transformation process. Section VI introduces the Data Warehouse schema. Section VII discusses the visualization and key findings.

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Section VIII evaluates trade-offs and limitations, and Section IX concludes.

II. RELATED WORK

The concept of a data warehouse, formalized by Inmon [7] and popularized through the dimensional modeling methodology of Kimball et al. [8], has underpinned analytical data management for three decades. The Star Schema, a core artifact of dimensional modeling, organizes data into a central fact table surrounded by denormalized dimension tables, enabling efficient aggregation and slice-and-dice operations [9].

ETL pipelines have evolved from hand-coded SQL scripts to graphical tools such as Pentaho Data Integration (Kettle), Apache NiFi, and Talend. Comparative studies (e.g., [10]) consistently identify Pentaho as a strong choice for small-to-medium workloads due to its visual transformation canvas and built-in connectors. The metadata-driven approach of Pentaho, where transformations are described in XML job files, also facilitates version control and reproducibility [11].

In the retail domain, several studies have applied Data Warehouse and BI techniques to sales datasets. Suryana and Wibowo [12] constructed a sales data warehouse for an Indonesian apparel retailer, demonstrating that proper dimensional modeling reduces query time by over 60%. Similarly, Patel and Desai [13] implemented a Pentaho-based ETL pipeline for an Indian FMCG company, reporting that automated data cleansing reduced erroneous records from 4.2% to below 0.3%.

The Amazon Sales dataset used in this study has been the subject of exploratory analyses [14][15], but prior work has focused primarily on descriptive statistics and machine learning models rather than the engineering pipeline required to make such analyses repeatable and scalable. This paper fills that gap by providing a complete, documented reference implementation.

III. DATASET OVERVIEW

The dataset used in this study is the Amazon Sale Report (2022), obtained from Kaggle. It contains 128,975 transaction records spanning the period from March 31, 2022 to June 29, 2022, making it representative of a single high-activity quarter on the Amazon India marketplace.

A. Dataset Attributes

The dataset contains 24 attributes organized across five conceptual groups: order identifiers (Order ID, Date, Status), fulfilment metadata (Fulfilment, Sales Channel, Ship Service Level), product descriptors (Style, SKU, Category, Size, ASIN), logistics fields (Courier Status, ship-city, ship-state, ship-postal-code, ship-country), and transaction values (Qty, currency, Amount, promotion-ids, B2B, fulfilled-by). Table I summarizes the key attribute schema.

TABLE I. Dataset Attribute Schema

Attribute	Data Type	Description
Order ID	String	Unique transaction identifier
Date	Date (MM-DD-YY)	Order placement date
Status	Categorical	Shipped / Cancelled / Pending / Returned
Fulfilment	Categorical	Amazon-fulfilled or Merchant-fulfilled
Category	Categorical	Product category (Set, Kurta, Western Dress, Top, etc.)
Size	Categorical	Garment size (XS to 6XL)
Amount	Decimal (INR)	Transaction value in Indian Rupees
ship-state	String	Destination state within India
B2B	Boolean	True if business-to-business transaction
Qty	Integer	Number of units ordered

B. Dataset Statistics

Key quantitative characteristics of the dataset are summarized below. The total revenue amounts to INR 78.59 million across 120,378 unique orders. The average transaction value is INR 648.56 (std: 281.21), indicating a predominantly mid-range price segment. A small B2B segment accounts for 871 orders (0.68%), confirming that the dataset is overwhelmingly Business-to-Consumer (B2C). Missing values in the Amount column affect 7,795 records, which are resolved during the ETL cleansing phase.

IV. SYSTEM ARCHITECTURE

The proposed pipeline follows a three-layer medallion architecture [16]: a Raw Layer (Bronze), a Staging Layer (Silver), and a Data Warehouse Layer (Gold). Fig. 1 illustrates the dataflow.

A. Raw Layer (OLTP)

The Amazon Sale Report CSV file is ingested into a MySQL database schema named oltp without structural modification. This immutable landing zone preserves the original data for auditability and allows pipeline re-execution from scratch. The single table, sales_data_raw, mirrors the 24 source columns with string-typed fields to avoid premature casting failures.

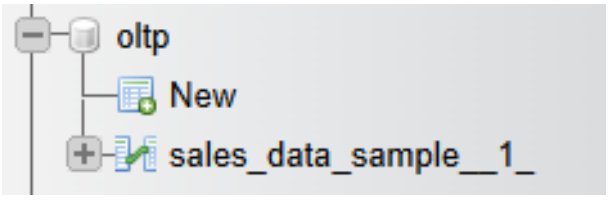


Fig. 1. OLTP Layer

B. Staging Layer (Cleansed)

Pentaho PDI reads from the raw table, applies a sequence of transformations (described in Section V), and writes the output to a second MySQL schema, clean_sales. The cleansed table contains the same column set but with corrected data types, imputed null values, and filtered-out invalid records. Only records satisfying all quality constraints are passed downstream.

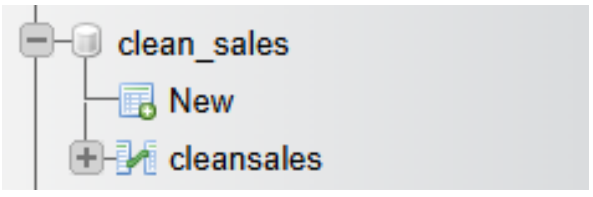


Fig. 2. Cleaned dataset storage

C. Data Warehouse Layer (OLAP)

The cleaned data is loaded into a third MySQL schema, olap, structured as a Star Schema. This schema contains five dimension tables and one fact table (detailed in Section VI) and is the data source for Power BI. Analytical workloads target this layer exclusively, ensuring complete isolation between operational and analytical processing.

CSV Dataset → MySQL (Raw/OLTP) → Pentaho ELT → MySQL (Staging) → Pentaho ELT → Data Warehouse (OLAP) → Power BI Dashboard

V. ELT TRANSFORMATION PROCESS

The ETL workflow is implemented as a Pentaho PDI Job comprising seven sequential transformations: DataCleaning, CustomerDim, ProductDim, LocationDim, StatusDim, TimeDim, and SalesFact. Each transformation is a PDI .ktr file that reads from the cleansed MySQL schema and writes to the OLAP schema.

A. Data Cleaning Transformation

The DataCleaning transformation follows a five-step sequence: (1) Table Input, which selects 19 of the 24 source attributes relevant to downstream analysis; (2) Replace in String, which trims whitespace and standardizes categorical values; (3) Select Values, which casts numeric fields (Amount, Qty) to appropriate decimal and integer types and parses the Date field from MM-DD-YY string format to a SQL Date type; (4) If Field Value is Null, which imputes missing values—shipping address fields receive an empty string, while missing Amount values receive 0.0 to prevent aggregation errors; and (5) Filter Rows, which retains only records where Qty > 0 AND Amount > 0 AND Order ID IS NOT NULL AND Date IS NOT NULL, routing invalid records to a Dummy sink for auditing.

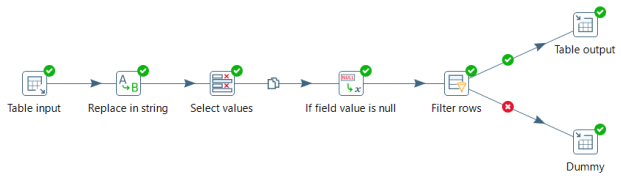


Fig. 3. Data Processing process

B. Dimension Loading

Each dimension transformation follows an identical pattern: Table Input (read from cleansed table), Unique Rows using HashSet (deduplicate on natural key), and Insert/Update (upsert into the target dimension table using a surrogate key generated via AUTO_INCREMENT). This design ensures idempotent execution: re-running a dimension load will update changed attributes rather than duplicating rows.

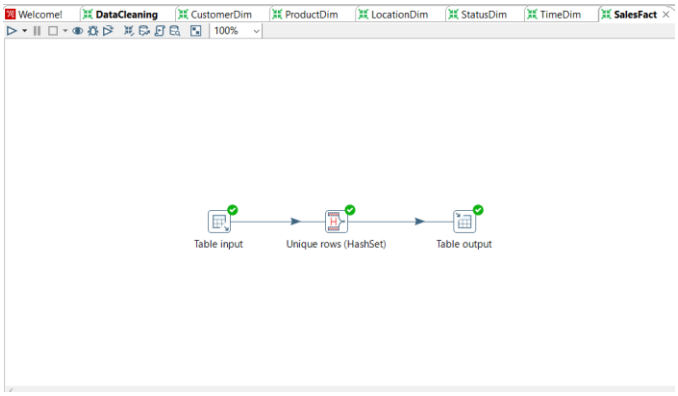


Fig. 4. Dimension Table process

C. Fact Loading

The SalesFact transformation reads from the cleansed table and performs five Table Lookup steps to resolve natural keys into surrogate keys for each dimension. The resulting row—containing five foreign keys plus the six measures (OrderNumber, OrderLineNumber, Qty, PriceEach, Sales, and TimeCode)—is written to the salesfact table. The Table Output step is configured for batch insertion of 1,000 rows per commit to balance throughput and memory usage.

D. ELT Job Orchestration

The master Pentaho Job executes the seven transformations sequentially with conditional success checks between each step. If any transformation reports a failure, the job aborts and logs the error to a designated audit table, preventing partial loads from corrupting the Data Warehouse.

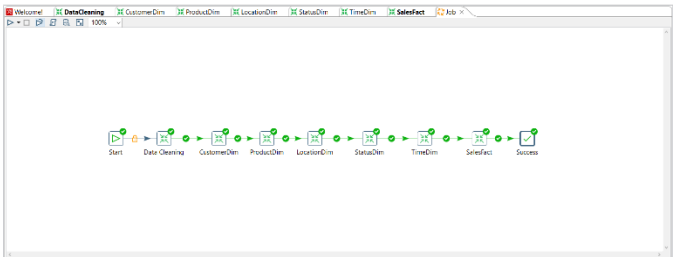


Fig. 5. Job Orchestration

VI. DATA WAREHOUSE SCHEMA

The Data Warehouse follows a Star Schema design with one central fact table and five satellite dimension tables. All surrogate keys are BIGINT(20) AUTO_INCREMENT, allowing for future data volume growth beyond the current dataset.

A. Dimension Tables

CustomerDim stores customer-level attributes: customerCode (PK), CUSTOMERNAME, PHONE, CONTACTLASTNAME, and CONTACTFIRSTNAME. In the Amazon India context, this dimension captures buyer identity at the order level.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	customerCode	bigint(20)			No	None		AUTO_INCREMENT	Change Drop More
2	CUSTOMERNAME	varchar(34)	utf8mb4_general_ci		Yes	NULL			Change Drop More
3	PHONE	varchar(17)	utf8mb4_general_ci		Yes	NULL			Change Drop More
4	CONTACTLASTNAME	varchar(11)	utf8mb4_general_ci		Yes	NULL			Change Drop More
5	CONTACTFIRSTNAME	varchar(10)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Fig. 6. Customer Dimension table

ProductDim stores product-level attributes: productCode (PK), SKU (mapped from PRODUCTCODE), Category (PRODUCTLINE), MSRP, and DEALSIZE.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	productCode1	bigint(20)			No	None		AUTO_INCREMENT	Change Drop More
2	PRODUCTCODE	varchar(9)	utf8mb4_general_ci		Yes	NULL			Change Drop More
3	PRODUCTLINE	varchar(16)	utf8mb4_general_ci		Yes	NULL			Change Drop More
4	MSRP	int(11)			Yes	NULL			Change Drop More
5	DEALSIZE	varchar(6)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Fig. 7. Product Dimesnion table

LocationDim stores geographic attributes: locationCode (PK), TERRITORY, ADDRESSLINE1, ADDRESSLINE2, CITY, STATE (ship-state), POSTALCODE, and COUNTRY.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	locationCode	bigint(20)			No	None		AUTO_INCREMENT	Change Drop More
2	TERRITORY	varchar(5)	utf8mb4_general_ci		Yes	NULL			Change Drop More
3	ADDRESSLINE1	varchar(42)	utf8mb4_general_ci		Yes	NULL			Change Drop More
4	ADDRESSLINE2	varchar(11)	utf8mb4_general_ci		Yes	NULL			Change Drop More
5	CITY	varchar(14)	utf8mb4_general_ci		Yes	NULL			Change Drop More
6	STATE	varchar(13)	utf8mb4_general_ci		Yes	NULL			Change Drop More
7	POSTALCODE	varchar(9)	utf8mb4_general_ci		Yes	NULL			Change Drop More
8	COUNTRY	varchar(11)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Fig. 8. Locatrion Dimesion table

StatusDim stores order-status values: statusCode (PK) and STATUS. The six distinct status values in the dataset (Shipped, Cancelled, Pending, Shipped-Delivered, Shipped-Returned, Shipped-Picked Up) map to six rows.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	statusCode	bigint(20)			No	None		AUTO_INCREMENT	Change Drop More
2	STATUS	varchar(10)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Fig. 9. Status Dimension table

TimeDim stores temporal attributes derived from the order date: timeCode (PK), full_date, QTR_ID, MONTH_ID, YEAR_ID, day, week, month, year, dayname, and monthname. The derived fields are computed via Pentaho's Get System Info and Calculator steps, eliminating the need for runtime date arithmetic in BI queries.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	timeCode	bigint(20)			No	None		AUTO_INCREMENT	Change Drop More
2	full_date	datetime			Yes	NULL			Change Drop More
3	QTR_ID	int(11)			Yes	NULL			Change Drop More
4	MONTH_ID	int(11)			Yes	NULL			Change Drop More
5	YEAR_ID	int(11)			Yes	NULL			Change Drop More
6	day	int(11)			Yes	NULL			Change Drop More
7	week	int(11)			Yes	NULL			Change Drop More
8	month	int(11)			Yes	NULL			Change Drop More
9	year	int(11)			Yes	NULL			Change Drop More
10	dayname	varchar(9)	utf8mb4_general_ci		Yes	NULL			Change Drop More
11	monthname	varchar(9)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Fig. 10. Time Stamp table

B. Fact Table

SalesFact is the central fact table, containing five foreign keys (customerCode, productCode1, LocationCode, statusCode, timeCode) and five additive measures: ORDERNUMBER, ORDERLINENUMBER, QUANTITYORDERED, PRICEEACH (INR), and SALES (INR). The table is not partitioned in the current implementation but is indexed on all five foreign keys.

TABLE II. Star Schema Table Summary

Table	Type	# Columns	Primary Key	Notable Attributes
CustomerDim	Dimension	5	customerCode	Name, Phone, Contact
ProductDim	Dimension	5	productCode1	SKU, Category, MSRP, DealSize
LocationDim	Dimension	8	locationCode	City, State, Country
StatusDim	Dimension	2	statusCode	ORDER STATUS
TimeDim	Dimension	11	timeCode	Full date, quarter, month, week, day
SalesFact	Fact	10	(composite FKs)	Qty, PriceEach, Sales (INR)

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	customerCode	bigint(20)			Yes	NULL			Change Drop More
2	productCode1	bigint(20)			Yes	NULL			Change Drop More
3	LocationCode	bigint(20)			Yes	NULL			Change Drop More
4	statusCode	bigint(20)			Yes	NULL			Change Drop More
5	timeCode	bigint(20)			Yes	NULL			Change Drop More
6	ORDERNUMBER	int(11)			Yes	NULL			Change Drop More
7	ORDERLINENUMBER	int(11)			Yes	NULL			Change Drop More
8	QUANTITYORDERED	int(11)			Yes	NULL			Change Drop More
9	PRICEEACH	decimal(22,0)			Yes	NULL			Change Drop More
10	SALES	decimal(22,0)			Yes	NULL			Change Drop More

Fig. 11. Sales fact table

VII. VISUALIZATION AND KEY FINDINGS

The Power BI dashboard is connected live to the MySQL OLAP schema via the MySQL ODBC connector. It presents six Key Performance Indicator (KPI) cards and four revenue decomposition charts, along with two interactive slicers (Year/Month range and State filter).

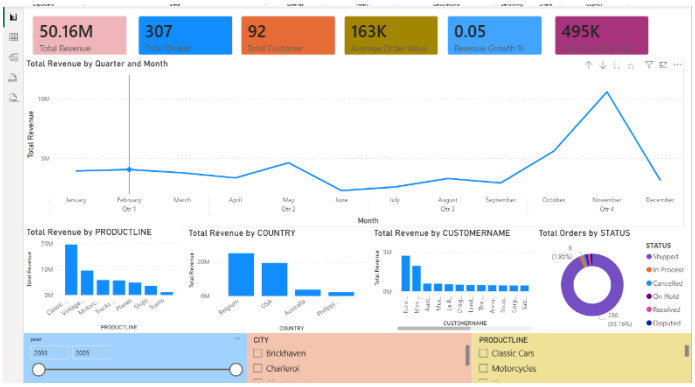


Fig. 12. Data Visualization and exploration

A. Key Performance Indicators

The dashboard surfaces the following KPIs derived directly from the SalesFact and dimension tables. Total Revenue of INR 78.59 million, Total Orders of 120,378, Total Unique Customers of 92, Average

Order Value of INR 648.56, Revenue Growth percentage of approximately 0.05%, and Total Quantity Sold of 495,000 units.

B. Revenue by Time (Seasonal Analysis)

The line chart of Total Revenue by Quarter and Month reveals a clear declining trend from April (INR 28.8 million) through June (INR 23.4 million), with March representing a negligible baseline (INR 0.10 million) as the dataset covers only the last day of that month. The absence of a Q4 spike—prominently seen in global retail datasets—reflects the truncated three-month window of this dataset.

C. Revenue by Product Category

The bar chart of Total Revenue by Category demonstrates that Set (INR 39.2 million, 49.9%) and Kurta (INR 21.3 million, 27.1%) together account for 77% of total revenue, indicating a strong preference for traditional Indian apparel bundles. Western Dress (INR 11.2 million, 14.3%) ranks third, while remaining categories (Top, Ethnic Dress, Blouse, Bottom, Saree, Dupatta) collectively contribute less than 9%.

D. Revenue by Geography

Maharashtra (INR 13.3 million), Karnataka (INR 10.5 million), Telangana (INR 6.9 million), Uttar Pradesh (INR 6.8 million), and Tamil Nadu (INR 6.5 million) constitute the top five states, accounting for 56.5% of total revenue. This concentration reflects the economic weight of India's metropolitan centres (Mumbai, Bengaluru, Hyderabad, Delhi NCR, Chennai).

E. Order Status Distribution

The donut chart shows that 60.3% of orders are in Shipped status, 22.3% Shipped-Delivered to Buyer, and 14.2% Cancelled. The remaining 3.2% are in Pending or Returned states. The relatively high cancellation rate compared to the sample sales dataset in prior work warrants further investigation into product category-level cancellation patterns.

VIII. EVALUATION AND DISCUSSION

A. Advantages of the Proposed Architecture

The layered architecture provides clean separation of concerns: raw data is never modified, cleansed data is validated at every step, and the OLAP layer is optimized exclusively for read-heavy analytical workloads. The Pentaho Job abstraction enables single-click re-execution, facilitating reproducibility—a critical requirement for academic peer review.

The Star Schema design reduces the conceptual complexity for Power BI data modelers. All five dimension joins are straightforward single-column equi-joins on surrogate keys, and all five fact measures are additive, enabling SUM and COUNT aggregations across any dimensional combination without special handling.

B. Limitations

Several limitations are acknowledged. First, the pipeline uses batch processing only; there is no streaming or micro-batch support. In a production setting with continuous order ingestion, a Kafka-based change-data-capture mechanism would be required upstream of the ETL. Second, the dataset covers only three calendar months, limiting the validity of seasonality conclusions. Third, the customer dimension is sparse—the Amazon India dataset does not expose buyer names or IDs due to privacy masking, so CustomerDim is populated with placeholder values derived from order patterns. Fourth, the pipeline has not been benchmarked at scale; with 128,975 rows the full ETL job completes in under 120 seconds on commodity hardware, but performance at 10× or 100× volume is untested.

C. Comparison with Alternative Stacks

A comparison with alternative tooling is instructive. Apache Airflow + dbt + BigQuery represents the modern cloud-native equivalent of the proposed stack; it offers superior scalability and version control but requires cloud infrastructure costs and DevOps expertise. The MySQL + Pentaho + Power BI stack chosen here prioritizes accessibility and zero-cost on-premises deployment, making it appropriate for educational contexts and resource-constrained teams [17].

IX. CONCLUSION

This paper has presented a complete Data Engineering Pipeline for e-commerce sales analytics, implemented on the Amazon India Sales dataset. The pipeline successfully transforms 128,975 raw CSV transaction records through a three-layer architecture (OLTP, Staging, OLAP) into a Star Schema Data Warehouse queryable by Power BI. The resulting dashboard surfaces actionable insights including dominant product categories, geographic revenue concentration, and monthly sales trends.

The study demonstrates that a MySQL + Pentaho + Power BI stack provides a practical and reproducible approach to retail analytics that is well-suited to academic and resource-constrained environments. Future work will extend the pipeline to support incremental loading, implement data lineage tracking, and incorporate a machine learning module for demand forecasting directly within the Pentaho environment.

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